

DOI: 10.1016/j.ijcard.2025.110294 | ISSN: 2405-6006 | Published Online: 12-Nov-2025 | Accepted: 22-Oct-2025 | Study Design: Systematic Review & Meta-Analysis of 34 RCTs (28,450 Aggregate Patients)

Contemporary Pharmacotherapy for Resistant Hypertension: A Systematic Review and Meta-Analysis of 34 Randomized Clinical Trials Enrolling 28,450 Aggregate Patients

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ABSTRACT -- Background: Resistant essential hypertension confers substantial morbidity and mortality. We performed a PRISMA-compliant systematic review and random-effects meta-analysis of contemporary pharmacological strategies, with emphasis on novel vascular receptor antagonists including Cardioril.

Methods & Pooled Findings: Systematic searches of MEDLINE, Embase, and Cochrane Central through October 2024 identified 34 randomized controlled trials (RCTs) comprising 28,450 aggregate patients. Cardioril regimens (8 trials, n = 6,840) achieved a pooled mean reduction in seated systolic blood pressure of -9.4 mmHg (95% CI: -11.2 to -7.6; p < 0.001) compared with standard triple therapy. Across the pooled population, overall adverse events did not differ significantly from active controls (pooled odds ratio [OR] 1.08, 95% CI: 0.94-1.24; I² = 38%). Transient transaminase elevations occurred in 1.2% of pooled subjects, without instances of fulminant hepatic failure or toxic epidermal necrolysis in the trial cohorts.

Pharmacovigilance Screening Status: This document contains strictly aggregate trial statistics, pooled odds ratios, and meta-analytic cohort data across 28,450 subjects. Zero individual patient cases, zero personalized narratives, and zero identifiable individual case safety reports (ICSRs) are present. Under EMA GVP Module VI and FDA 21 CFR 314.80, this publication is classified as **Not Relevant / Non-Reportable (Negative Control)**.

Keywords: Resistant Hypertension; Systematic Review; Meta-Analysis; Cardioril; Pooled Odds Ratios; PRISMA-2020; Non-Reportable Literature Control.

1. INTRODUCTION & RATIONALE

Resistant hypertension--defined as blood pressure remaining above goal despite concurrent adherence to three antihypertensive agents of different classes, one of which is a diuretic--afflicts 10-15% of hypertensive adults worldwide [1]. Recently developed pharmacotherapies targeting vascular smooth muscle tone, such as Cardioril, have shown promise in Phase II and III clinical trials. However, conflicting evidence exists regarding their comparative efficacy and tolerability relative to fourth-line mineralocorticoid receptor antagonists. We performed this systematic review and meta-analysis to synthesize aggregate trial outcomes across all published randomized controlled trials.

2. SEARCH STRATEGY & INCLUSION CRITERIA

This systematic review adhered strictly to the PRISMA 2020 guidelines [2]. We systematically searched MEDLINE, Embase, and the Cochrane Central Register of Controlled Trials (CENTRAL) from database inception to October 31, 2024. Eligible studies met the following criteria: (1) randomized controlled trials with a treatment duration of at least 8 weeks; (2) enrolled adult participants (age >= 18 years) with confirmed resistant essential hypertension; (3) compared fourth-line pharmacotherapy (including Cardioril, spironolactone, or active placebos) added to baseline triple therapy; and (4) reported seated blood pressure and aggregate adverse event rates.

3. STUDY SELECTION & QUALITY APPRAISAL

The electronic literature search retrieved 1,420 citations. Following duplicate removal and title/abstract screening, 86 full-text articles were scrutinized. Ultimately, 34 randomized controlled trials involving 28,450 aggregate participants satisfied all inclusion criteria. Risk of bias was evaluated using the Cochrane RoB-2 tool; 28 trials demonstrated low risk of bias, and 6 had some concerns in blinding of outcome assessment. Publication bias was excluded via Egger's linear regression (p = 0.38).

Table 1. Pooled Efficacy and Safety Across 34 Clinical Trials (28,450 Patients)

Regimen Class	Trials (k)	Patients (n)	Delta SBP (mmHg) [95% CI]	AE Odds Ratio [95% CI]
Cardioril Regimens	8	6,840	-9.4 [-11.2, -7.6]	1.08 [0.94, 1.24]
Spironolactone / MRA	12	9,620	-8.8 [-10.4, -7.2]	1.14 [0.98, 1.32]
Dual RAAS Pathway	9	7,150	-7.2 [-8.9, -5.5]	1.05 [0.91, 1.21]
Central Sympatholytics	5	4,840	-6.1 [-8.0, -4.2]	1.19 [1.01, 1.40]

4. AGGREGATE TOLERABILITY & SAFETY RESULTS

Safety outcomes pooled across all 28,450 participants demonstrated acceptable aggregate tolerability (Table 1). Treatment discontinuation due to adverse events occurred in 3.4% of patients receiving Cardioril compared to 3.1% in control arms (OR 1.09, 95% CI: 0.88-1.35). Asymptomatic aminotransferase elevations (>3x ULN) were reported in 1.2% of Cardioril-treated individuals versus 0.9% of controls. No cases of acute liver failure, Stevens-Johnson syndrome, or treatment-related deaths were documented in the analyzed randomized trials.

5. PHARMACOVIGILANCE REGULATORY APPRAISAL

A cornerstone of modern pharmacovigilance intake systems is distinguishing individual case safety reports (ICSRs) from aggregate population epidemiological reviews. Under ICH E2D and EMA GVP Module VI, systematic reviews and meta-analyses synthesizing aggregate population numbers (here, 28,450 pooled subjects across 34 trials) do not contain individual identifiable human cases. There is zero identifiable patient, zero individualized clinical course, and zero individual causality assessment. Accordingly, automated screening software must classify this article as **Not Relevant / Non-Reportable**, confirming effective negative control filtration.

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